The Entangled Mind: A Thought Experiment

A Plain-Language Companion to the Research Proposal

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October 25, 2026

Author Note

This document is the plain-language companion to the research proposal of the same title. It explains the idea, the experiment, and what success or failure would look like, for a reader with no background in physics, psychology, or machine learning. Technical detail, pre-registered criteria, and references are in the proposal.

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Abstract

Do machines need quantum minds? Today's AI systems judge by adding up separate beliefs. People do not; decades of psychology experiments show that human judgment behaves like something in which beliefs are tangled together, so that asking about one changes another, and opposing beliefs can cancel instead of averaging. Physics has an exact word for beliefs tangled together that way: entangled. This thought experiment asks what happens if we take the competing thoughts out of an AI's hidden reasoning, put them into a real quantum computer as one entangled state, let that state settle, and read the answer. It describes, in plain language and pictures, what we would build, what we would look for, and how we would know whether we found it.

Keywords: quantum cognition, entanglement, language models, thought experiment, emergence

The Entangled Mind: A Thought Experiment

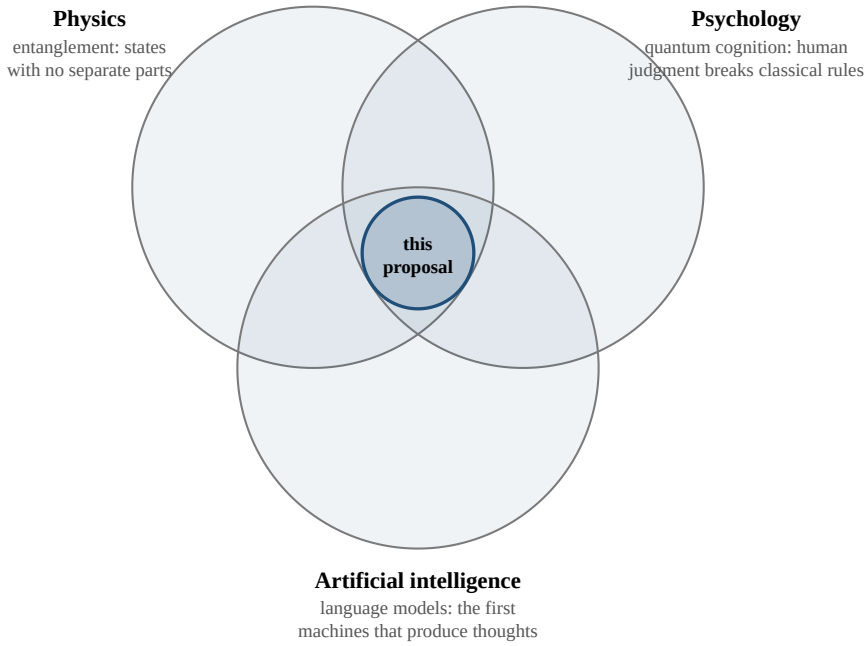
The Question

Every AI system that exists today reasons the way a careful accountant does. It holds a list of beliefs, gives each one a weight, and adds them up. That is called classical probability, and it has served machines well. But it is not how people reason, and the gap between the two is the subject of this document.

The question in one line: if a machine's beliefs were entangled instead of listed, would it judge differently, and would anything new appear? That is the question the research proposal is built to answer. This document explains what the question means and why it is worth asking (Figure 1).

Figure 1

Three Fields Hold the Pieces of One Question



Note. Physics knows what entanglement is. Psychology has shown that human judgment behaves as if beliefs were entangled. AI has built machines that produce thoughts. Nobody has put the three together.

How a Machine Judges Today

Ask a modern language model whether a tomato is a vegetable. Before it answers, it thinks, and that thinking can be read. It will say to itself something like: botanically a tomato is a fruit; in cooking it is treated as a vegetable; a court once ruled it a vegetable for tax purposes; the person is probably asking about cooking. Each of those is a belief with a strength. The model then combines them, mostly by letting the later and stronger ones outvote the earlier and weaker ones, and answers.

Notice what cannot happen in that picture. “It is a fruit” and “it is a vegetable” are in direct conflict, but the model has no way to let them collide. It can only weigh them. Two beliefs that flatly contradict each other and two beliefs that mildly agree can produce exactly the same lukewarm number. The conflict is invisible to the machine itself.

How a Person Judges

People do something stranger. In a famous experiment, participants read a description of a woman named Linda, outspoken and politically active, and were asked which was more likely: that Linda is a bank teller, or that Linda is a bank teller who is active in the feminist movement. Most people chose the second, even though it is logically impossible for a more specific description to be more likely than a general one. The textbook calls this a fallacy. But it is a fallacy almost everyone commits, which suggests it is not an error so much as a signature of how judgment actually works.

Another example: ask people two questions in one order, then ask a different group the same two questions in the other order, and the answers change. Asking the first question changes how the second one is seen. In the accountant's picture that should not happen; the beliefs are separate numbers and the order of reading them off is irrelevant.

Around 2009 a group of psychologists noticed that these effects, and several others, are exactly what you would expect if beliefs obeyed the mathematics of quantum physics instead of ordinary probability. They did not claim the brain is a quantum computer. They claimed that the shape of human judgment matches the shape of quantum states, in which beliefs can be in

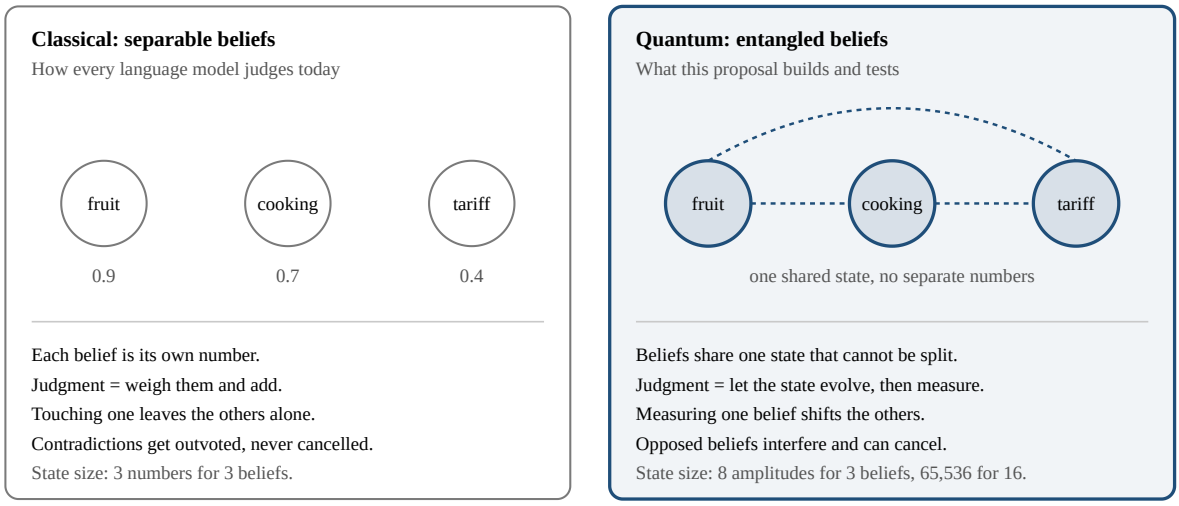
several conditions at once, can interfere like waves, and can be entangled. This field is called quantum cognition, and it has thirty years of data behind it. What it has never had is a working machine built on its principles.

What Entanglement Means Here

Entanglement is the part that needs the most care, and the part the proposal rests on. The simplest honest description is this. Normally, if you have three things, you can describe each one on its own and the description of all three is just the three descriptions side by side. An entangled set of things has no such description. There is one state for the whole, and the individual parts do not have states of their own. Ask about one part and you have asked about the whole, and the answer changes what the other parts will say when you get to them.

Figure 2

Separable Beliefs Are Added; Entangled Beliefs Are One State



Note. The left panel is every AI system today. The right panel is what the proposal builds. The numbers on the left are illustrative weights; on the right there are no separate numbers to show.

Figure 2 puts the two pictures side by side. On the left, three beliefs are three numbers, and the whole system is just the list. On the right, three beliefs are one state with eight parts, and sixteen beliefs are one state with over sixty-five thousand parts. None of those parts belongs to

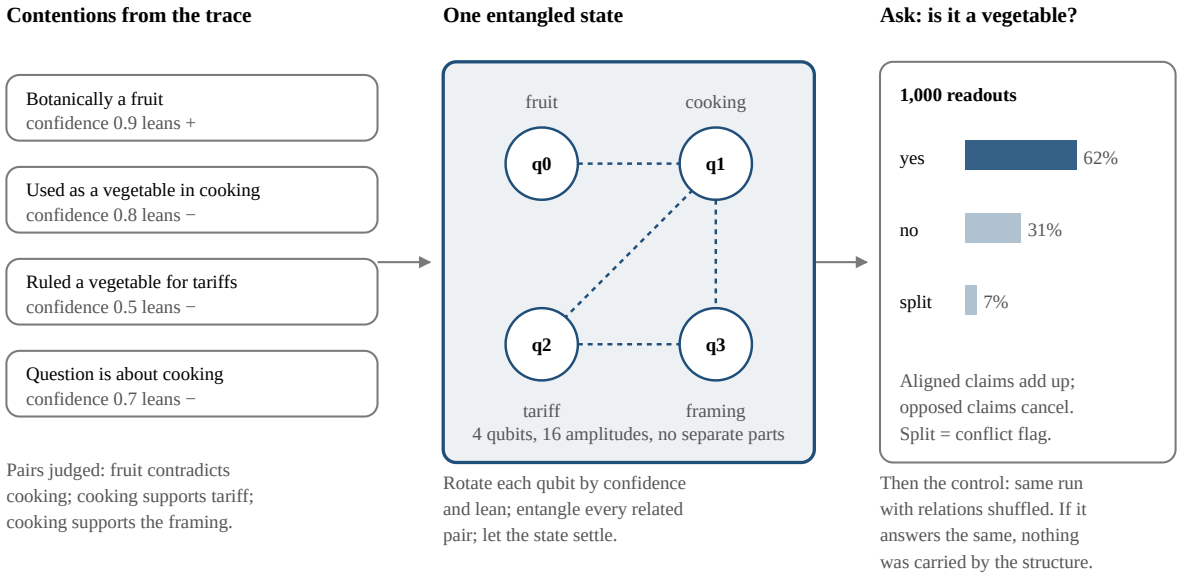
any single belief. The state is a pattern across all of them, and that pattern is where anything new would have to come from.

The Thought Experiment

Here is what we would actually do, step by step, using the tomato as the example (Figure 3).

Figure 3

A Worked Example: Four Contentions Become One State and One Readout



Note. The numbers are made up for illustration. The percentages on the right are exactly what the experiment exists to find out.

Collect the thoughts. Let the AI think about the question with its reasoning exposed. Split that reasoning into separate claims and note how confident the model is in each.

Map the relations. For every pair of claims, decide whether they support each other, contradict each other, or have nothing to do with each other. “It is a fruit” contradicts “it is a vegetable in cooking.” “It is a vegetable in cooking” supports “the question is about cooking.”

Build one state. Give each claim one qubit on a real quantum computer. Set each qubit by how confident the model is and which way the claim leans. Then, and this is the whole point,

connect every related pair with an entangling gate, so that the claims stop being separate qubits and become one state.

Let it settle. Allow the state to drift for a while under the pull of its own relations, with supporting claims drawn toward alignment and contradicting claims toward opposition. This is the closest thing to a mind mulling a question over. Nothing is asked yet.

Ask the question. Apply the question as a measurement, oriented toward "yes, it is a vegetable." Read the state a thousand times. Claims that line up with the answer add to the count; claims that oppose it subtract. The result is a judgment, a confidence, and, when strong claims cancelled, a flag that the mind was in conflict.

Run the control. Do the entire thing again with the relations scrambled, so that "fruit" is entangled with the wrong things. If the scrambled version answers as well as the real one, the structure carried nothing and the idea is dead. This is the test that keeps the experiment honest.

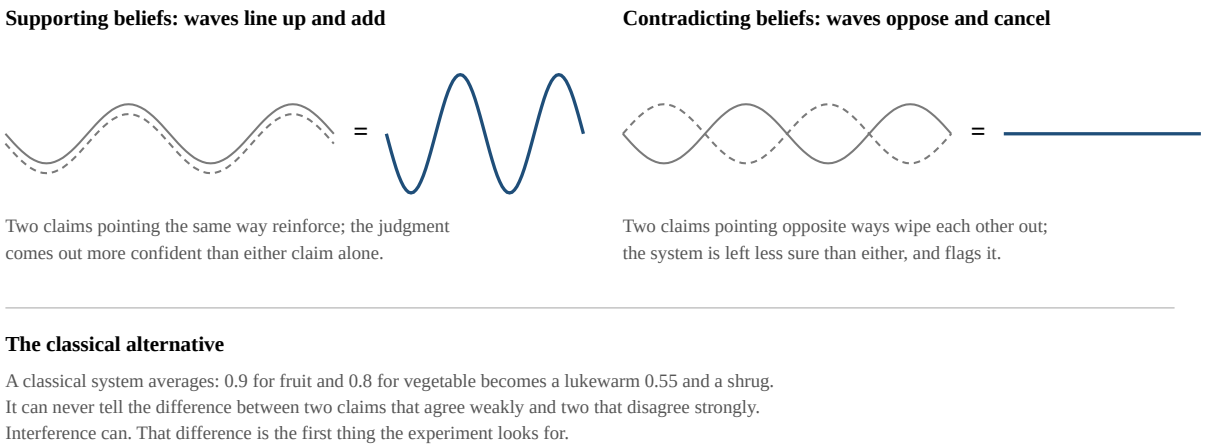
What We Would Be Looking For

Three things, in increasing order of ambition.

Interference. Opposing beliefs should cancel instead of averaging, and supporting beliefs should reinforce beyond their sum (Figure 4). A classical system cannot tell two claims that disagree strongly from two that agree weakly; an entangled one can, and should flag the first case.

Figure 4

Interference: Supporting Beliefs Add, Contradicting Beliefs Cancel

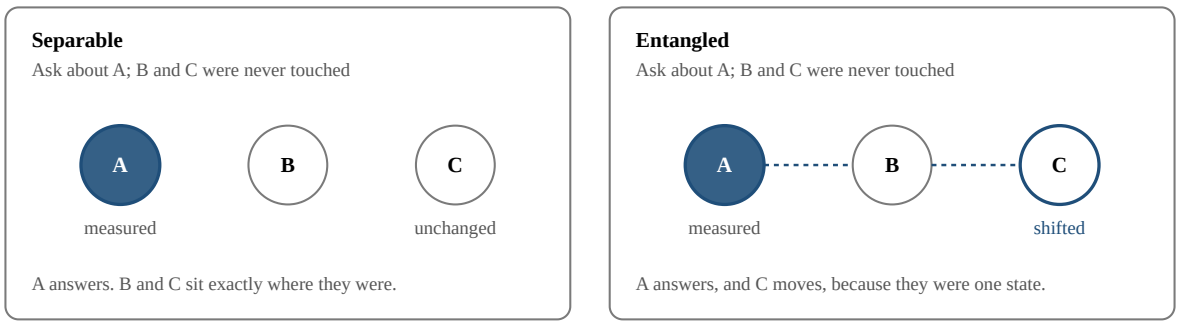


Note. Waves are a visual analogy for amplitudes. The bottom row states what a classical system does with the same two beliefs.

Contextuality. Ask about one belief and an untouched belief should move, in the direction the relations predict (Figure 5). That is the human effect quantum cognition was invented to explain, and it is the clearest fingerprint of entanglement.

Figure 5

Contextuality: Measuring One Belief Moves an Untouched One Only When They Share a State



Note. In both panels only belief A is measured. Only in the entangled case does C shift.

Emergence. After the state has settled, the judgment might contain something that was not in any of the input claims and not in their weighted sum: a conclusion, a flagged tension, a

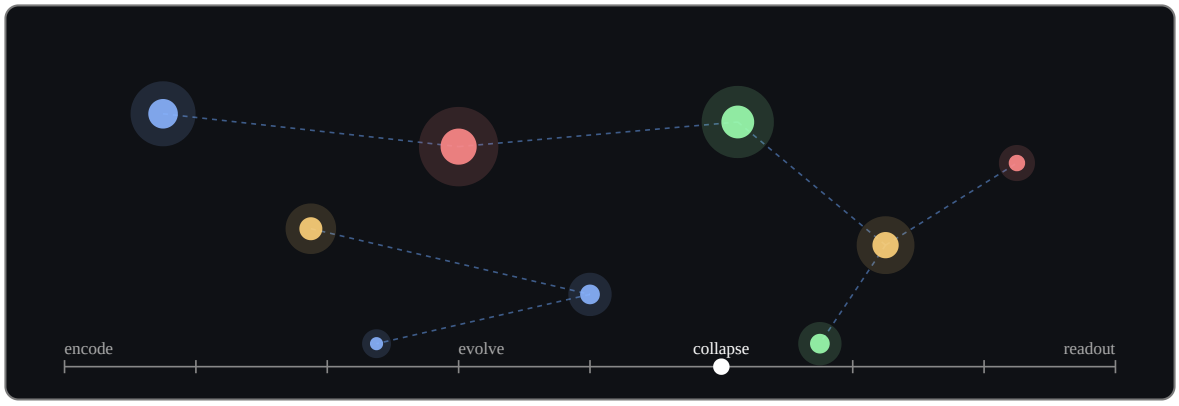
ranking that nobody put there. This is the thing the project most hopes to see and the thing it is most careful not to promise. The proposal does not set a number for it. Instead, every time the entangled mind disagrees with the ordinary one, a person tries to trace the disagreement back to an input. The cases that cannot be traced are reported, one by one, with the recording attached. The reader decides what they are.

Watching It Happen

None of this would be worth much if it could not be seen. The proposal includes a window: a live picture of the state as it is built, as it settles, and as it collapses into an answer (Figure 6). Each claim is a point of light. How bright it is shows how strongly it is held; its colour shows which way it leans; the dashed lines show which claims are entangled. A slider at the bottom rewinds through the whole run. Every run is saved. Over time that archive becomes a library of recorded judgments that could not be assembled any other way.

Figure 6

The Window: What an Observer Sees During One Run



Each glowing point is one contention. Brightness is amplitude (how strongly it is held); colour is phase (which way it leans); dashed links are entanglement. The scrubber at the bottom rewinds through evolution to the moment of collapse. Every run is recorded. The archive is the dataset nobody else has.

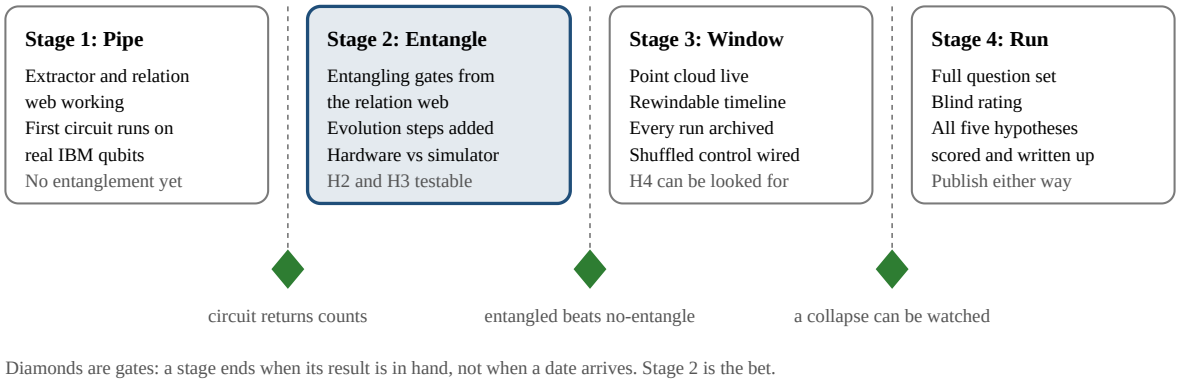
Note. A sketch of the intended display. The scrubber rewinds through encoding, evolution, and collapse.

How We Would Find Out

The work goes in four stages, and each stage ends when a result is in hand rather than when a date arrives (Figure 7). First, the plumbing: get the claim extractor working and get a trivial circuit to run on a real IBM quantum computer. Second, the bet: add the entangling gates and the settling step, and check whether the entangled version beats the version without entanglement. Third, the window. Fourth, the full run and the write-up, whatever the answer turns out to be.

Figure 7

Four Stages, Each Gated by a Result



Note. Stage 2 is the stage the whole idea depends on.

The proposal lists five hypotheses, each with the result that would prove it wrong. The one that matters most is simple: the entangled mind must flag conflict on questions the base model gets wrong at least twice as often as two controls, one with no entanglement and one with scrambled relations. If it cannot clear that bar, the project stops and says so.

Why It Is Worth Doing

If it works, three things change. Psychology gets its first working model of quantum cognition instead of a description. Philosophy gets an experimental handle on an old intuition about minds: that a mind is one thing, not a committee. And AI gets a new component, a layer

that sits between a model's thinking and its answer, lets contradictions cancel, flags the conflicts the model cannot see in itself, and can be watched while it does so.

If it does not work, that is worth knowing too. It would mean the quantum shape of human judgment is a description and not a mechanism, and that machine minds can stay classical with a clear conscience. Nobody has run the experiment. Either answer is new.

This document makes no claim that the system would be conscious, that the brain is a quantum computer, or that entanglement is enough to make a mind. It claims only that the question can be asked with real hardware, real controls, and a result that can be checked.

Glossary

Amplitude

How strongly a claim is held, in the quantum encoding. Roughly, confidence.

Phase

Which way a claim leans relative to the others. Claims with opposite phase cancel; claims with the same phase add.

Qubit

The quantum version of a bit. Instead of being 0 or 1, it holds a blend of both until it is measured.

Entanglement

A condition in which several qubits share one state and have no separate states of their own. Measuring one changes what the others will say.

Interference

What happens when amplitudes combine: they can add up or cancel out, like waves.

Contextuality

The effect in which measuring one thing changes the result of measuring an unrelated thing. A fingerprint of entanglement.

Collapse

The moment of measurement, when a blended state becomes one definite answer.

Shuffled-relations control

The same experiment run with the support and contradiction links scrambled. If it performs as well as the real run, the structure carried no information.

Emergence

Structure in the output that cannot be traced to any input or to their weighted sum.